A cautionary note on word learning tasks and presupposition triggering

Johanna Alstott[1*] and Athulya Aravind[2]

[1*]Massachusetts Institute of Technology.
[2]Yale University.

*Corresponding author(s). E-mail(s): jalstott@mit.edu;
Contributing authors: athulya.aravind@yale.edu;

Abstract

Cross-linguistically, predicates with initial-state and change-of-state components tend to encode them as presupposition and assertion, respectively. Bade, Schlenker, and Chemla (Natural Language Semantics 32(4):473–503, 2024)–henceforth BSC24–argue based on a series of artificial word learning experiments that this cross-linguistic tendency reflects conceptual biases privileging changes-of-state over initial-states. In their experiments, they gauged how participants encoded the initial-state and change-of-state entailments of a nonce verb *wug*, and they interpret their results as suggesting that participants generally encoded the initial-state entailment as a presupposition and encoded the change-of-state entailment as the assertion. This finding, they argue, favors their conceptual-bias hypothesis over competing accounts. In this paper, we further test the validity of BSC24's paradigm via an additional experiment where we both try to replicate their original effect and, in parallel, ascertain whether their results generalize to a nonce initial-state/change-of-state predicate other than the one they test. We not only fail to extend BSC24's results to our new nonce word but also fail to replicate their original effect: our participants overwhelmingly treated BSC24's *wug* and our new nonce word as non-presuppositional. Furthermore, a closer look at BSC24's original studies suggests that non-presuppositional construals were common there, too. We discuss several reasons why this could have been the case. All told, our outlook is pessimistic: adult artificial word-learning tasks do not, in fact, illuminate the mechanisms of presupposition triggering.

Keywords: presuppositions; word learning; change-of-state verbs; triggering

1 Introduction

Cross-linguistically, when a predicate has both an initial-state entailment and a change-of-state entailment, the initial-state entailment gets encoded as a presupposition, while the change-of-state entailment gets encoded as the assertion. As an example, consider the English verb *stop*, as used in (1). (1) has both the initial-state entailment in (1a) and the change-of-state entailment in (1b), and under standard analyses, the former entailment is a presupposition and the latter is an assertion.

(1) Mary stopped smoking.
  a. Initial-state entailment: Mary used to smoke.
  b. Change-of-state entailment: Mary does not smoke now.

Predicates that reverse the division of labor between presupposition and assertion–that is, predicates that assert an initial state and presuppose a change-of-state–are unattested, or at least substantially less common. For example, there is no verb that, when placed in the sentence frame "Mary [verb]-ed smoking," asserts (1a) and presupposes (1b), despite the fact that one can write a lexical entry for such a verb. Predicates that assert both an initial state and a change-of-state are similarly rare.

Following Abrusán (2010, 2011) and Schlenker (2021), Bade, Schlenker, and Chemla 2024 (BSC24) hypothesize that initial-state and change-of-state entailments tend to be presupposed and asserted, respectively, because of conceptual biases privileging changes-of-state over initial states. In other words, BSC24 propose that humans encode change-of-state events in such a way that the initial state is backgrounded, with the result that initial-states and changes-of-state are linguistically represented as presuppositions and assertions, respectively. This conceptual-bias view contrasts with what BSC24 call "contextual" approaches, according to which predicates like *stop* presuppose an initial state and assert a change-of-state because of how *stop* relates to its lexical alternatives (Abusch 2002, 2010; Romoli 2011) or the implicit QUD (Simons et al. 2010, 2017; Tonhauser et al. 2018).

Seeking to assess their conceptual-bias hypothesis experimentally, BSC24 reason that if the relevant conceptual biases exist, someone who is taught a nonce word with an initial-state entailment and a change-of-state entailment should (as a result of their conceptual biases) spontaneously encode the initial state as a presupposition and the change-of-state as the assertion. "Contextual" approaches do not share this prediction, BSC24 argue, because nonce words do not have lexical alternatives or a well-defined information status.

To test this prediction, BSC24 ran a two-phase artificial word learning experiment. During their training phase, participants saw four animations of a green circle moving upwards from a line, and participants were told that *the green circle wugs* describes all four animations. This training was intended to teach participants that *the green circle wugs* has the initial-state entailment in (2a) and the change-of-state entailment in (2b).

(2) The green circle wugs.
  a. Initial-state entailment: The green circle's initial position is on the line.
  b. Change-of-state entailment: The green circle moves upward.

During the test phase, BSC24 gauged how participants encoded the entailments of *wug* using a covered box task (Huang et al. 2013). On each trial, participants saw a sentence and two animations (a visible one and a "hidden" one), and they had to guess which animation the sentence described. BSC24 had a variety of trial types, but the most important trials were ones whose visible animation falsified (2a) but verified (2b) (i.e. a green circle moving upward, but not from the line). On these trials, BSC24's participants said that *the green circle wugs* describes the visible animation (rather than the covered box) roughly 50% of the time, and they said at a similar rate that *the green circle does not wug* describes the visible animation. In covered-box tasks, this absence of a negation contrast is the pattern that surfaces when the visible animation falsifies a presupposition (see BSC24 for overview of relevant literature), and hence, they conclude that participants encoded the initial state as a presupposition.

Conversely, on trials whose visible animation falsified the change-of-state component of (2) rather than the initial-state component (i.e. a circle moving downward from the line), there was a negation contrast: participants said ~80% of the time that *the green circle does not wug* describes the visible animation, while they said that *the green circle wugs* describes the visible animation ~35% of the time. This negation contrast suggests that participants treated (2) as false (rather than undefined) when (2b) is not met: that is, they encoded (2b) as an assertion.

BSC24 interpret the results of this experiment (their Experiment 1) as favoring the conceptual-bias hypothesis. After Experiment 1, BSC24 pressure-test their paradigm via two more experiments, both of which use the meaning for *wug* given in (2). In Experiment 2, BSC24 find that the results of Experiment 1 replicate even when the training phase biases participants against a presuppositional construal for (2a); however, they curiously also find that Experiment 1's results do not replicate when the training phase is biased *in favor of* a presuppositional construal, meaning that the results of Experiment 2 are mixed. Finally, Experiment 3 shows that quantificational sentences with *wug* exhibit the projection behavior we would expect if (2a) is a presupposition. Considered collectively, BSC24 argue, their results are in line with the conceptual-bias view.

Our aim in this paper is to further test the validity of BSC24's word-learning paradigm for making inferences about presuppositions, and in turn, test the validity of their conclusions. To that end, we ask whether BSC24's results **generalize** to other nonce change-of-state predicates beyond the one that they test. For example, do their results persist in a variant of their experiment where *wug* has, e.g., color-related initial-state/change-of-state entailments rather than spatial ones? Since BSC24's hypothesis concerns not just the encoding of spatial initial-states and changes-of-state (e.g. (2a-b)) but rather the encoding of initial-states and changes-of-state in general, they crucially predict that their results should generalize in this way. Until this prediction has been tested, it remains unknown whether BSC24's results are really teaching us something about the encoding of initial-states and changes-of-state in general, or whether their results apply only narrowly to how humans encode spatial words/events.

To address the question of how generalizable BSC24's results are, we undertake an additional experiment where we both try to replicate their Experiment 1 and, in parallel, run a version of their Experiment 1 that uses a color-based meaning for *wug*. In particular, *wug* in our color-based experiment will have the initial-state entailment in (3a) and the change-of-state entailment in (3b).

(3) The square wugs.

a. <u>Initial-state entailment</u>: The square's initial color is grey.

b. Change-of-state entailment: The square darkens.

As detailed in section 2.1, the logic and design of our color-based experiment is fully parallel to BSC24's study, and hence, they predict that their results should extend to this variant. In particular, if there is a general conceptual bias to treat initial-states as presupposed and changes-of-state as asserted (such as the biases proposed by Abrusán (2010) and Schlenker (2021)), then participants should not only construe (2a) and (2b) as the presupposition and assertion (respectively) of (2), but they should construe (3a) and (3b) as the presupposition and assertion of (3).[1]

Contrary to this prediction, we not only fail to extend BSC24's results to our color-based *wug* but also fail to replicate their original effect. In other words, the vast majority of our participants did not adopt presuppositional construals of (2) or (3). While several kinds of non-presuppositional construals were common in our experiment, the most common by far were what we call "ignore initial state" construals: most of our participants did not treat *wug* as having an initial-state entailment at all, interpreting (2) as simply "moves up" and interpreting (3) as simply "darkens."

As BSC24 themselves state, "if [ignore initial state construals] were generally available, it would be problematic for the paradigm we suggest here" (p. 490). In light of our finding that "ignore initial state" construals appear to indeed be generally available, the question is why our results (seemingly) differed from those of BSC24. A closer look at BSC24's Supplementary Materials shows that despite appearances, a substantial proportion of BSC24's participants–up to half–likely adopted non-presuppositional construals of *wug*. The high rate of non-presuppositional construals was not always visible in BSC24's experiments because their "ignore initial state" participants, unlike ours, were counter-balanced by participants who adopted another type of non-presuppositional construal: a conjunctive construal (*wug* as asserting (2a) and (2b)). The upshot is that non-presuppositional construals were likely widespread not just in our experiment but in theirs.

Having established the motivation for our experiment and previewed our results, we now lay out the methodology and results of our experiment in detail.

2 Experiment

Our experiment was pre-registered via the Open Science Foundation (OSF). For links to our pre-registration, stimuli, data, and analysis script, see "Data Availability," below.

[1] To elaborate: while one could claim that there are different conceptual biases for space vs. color, the theories that BSC24 take their results to support do not make such a distinction. For example, under Abrusán's (2010; 2011) theory, what explains the cross-linguistic tendency in the encoding of initial states and changes-of-state is a temporal bias, namely a bias that favors presupposing information that is not about the topic time (e.g. initial-state entailments) and asserting information that is about the topic time (e.g. change-of-state entailments). For Schlenker (2021), what explains the cross-linguistic tendency is a bias to presuppose information that is likely to be antecedently believed; initial states are more likely to be antecedently believed than changes-of-state (since initial states are further in the past), and as such they are more likely to be presupposed. Given how general these biases are, they should play a role not just in participants' encoding of (2) but also in their encoding of (3).

2.1 Methodology

2.1.1 Participants

Our study had two between-subjects conditions: the spatial condition and the color condition. BSC24 found significant effects in their Experiment 1 with 49 participants; rounding this up to 50, we aimed to have 50 participants per condition in our analyzed sample. To achieve this goal, we started by recuiting an initial 50 participants per condition, excluded those participants who failed to meet our criteria for inclusion in analysis, and kept recruiting participants until we had 50 participants per condition who met our inclusion criteria. Our participants, like BSC24's were recruited via Prolific; each of them was paid £1.20 for our roughly 8-minute study, regardless of whether they were included in our analyzed sample.

In total, we recruited 69 participants in the spatial condition and 63 participants in the color condition. 16 participants in the spatial condition and 9 participants in the color condition were excluded due to a technical issue where the animations either did not load properly or took a very long time to load. Only a handful of participants explicitly messaged us about loading issues of this sort, but all of those who did had a "_PreloadFailed_" warning in their data; hence, we opted to exclude all participants with such a warning. We believe that the issue was likely caused by server overload (i.e. too many participants taking the experiment at once), as evidenced by the fact that they problem started to go away once we tried recruiting in smaller batches.

In addition to these exclusions due to technical error, we also implemented the attention check used in BSC24's Experiment 1, i.e. we excluded participants who responded incorrectly on both of the test trials whose animations were just like training.[2] 3 participants in the spatial condition and 4 participants in the color condition were excluded under this criterion. Thus, we had 50 participants in each condition's analyzed sample, as desired.

2.1.2 Materials

The spatial condition was a straight replication of BSC24's Experiment 1. In the training phase, participants saw four animations of a green circle moving upwards from a red line, and participants were told that *the circle wugs* describes all four animations.[3] As in BSC24's Experiment 1, the position of the circle on the line and the distance that it moved varied across the four animations. This variation helps discourage participants from assuming that *wug* has entailments about the exact position of the circle on the line or about how far the circle moves. Participants could play the training (and test) animations as many times as they liked. However, if a participant

[2] To use the terminology defined below in section 2.1.3, this amounted to (a) excluding participants in the spatial condition who chose the covered box in their positive "StartsRed$_{True}$, MovesUp$_{True}$" trial and chose the visible animation in their negative "StartsRed$_{True}$, MovesUp$_{True}$" trial; (b) excluding participants in the color condition who chose the covered box in their positive "StartsGrey$_{True}$, Darkens$_{True}$" trial and chose the visible animation in their negative "StartsGrey$_{True}$, Darkens$_{True}$" trial.

[3] Our use of *the circle* rather than *the green circle* constitutes a minor change from BSC24. They presumably used *the green circle* to ensure that participants focused on the green circle in their Figure 1 (p. 480) rather than the large grey circle in which the green circle moves. Although *the circle* is technically ambiguous in this way, our results suggest that *the circle* was interpreted in the intended way–see Figure 7.

tried to progress to the next trial without viewing their current trial's animation at least once, our experimental script prevented them from doing so.

During the test phase of the spatial condition, we assessed participants' encoding of *wug* using a covered-box task, in the same manner as BSC24: on each trial, participants saw a sentence and two animations (a visible one and a "covered" one), and their task was to guess which one the sentence described (see Figure 1).

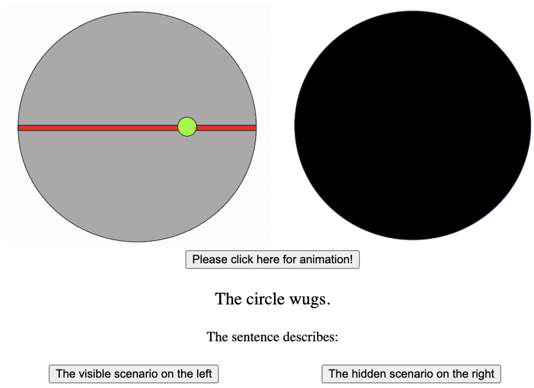


Fig. 1: Screen shot from the spatial condition's test phase.

As in BSC24's Experiment 1, our critical trials used four types of visible animations that varied along two dimensions: "StartsRed" (i.e. whether the circle in the animation starts on the red line) and "MovesUp" (i.e. whether the circle in the animation moves upward).[4] On "StartsRed$_{True}$, MovesUp$_{True}$" trials (which BSC24 call "True Controls"), the visible animation depicted a green circle moving upwards from the line (as in training). On "StartsRed$_{True}$, MovesUp$_{False}$" trials (which they call "Target A"), the visible animation depicted a green circle moving downward from the line. On "StartsRed$_{False}$, MovesUp$_{True}$" trials (which they call "Target B"), the visible animation showed a green circle moving upward, but not from the line. Finally, on "StartsRed$_{False}$, MovesUp$_{False}$" trials (which they call "False Controls"), the visible animation showed a green circle moving downwards and not from the line.

All animation types occurred with positive sentences ("the circle wugs") in half of trials and with negation ("the circle does not wug") in the other half, leading to a 2x2x2 design (MovesUp x StartsRed x Polarity) with eight trial types (Table 1).

In addition to critical trials, participants in the spatial condition also saw quantified fillers ("Every/none of the circles wug"), following BSC24. In particular, participants saw exemplars of each of the trial types shown in BSC24's Table 5, p. 493. As shown in that table, half of the fillers used "every" and half used "none of the"; additionally, the fillers' visible animations depicted four circles rather than one and showed a variety of starting positions and movement directions.

[4] BSC24 treat "animation type" as a four-level factor, but we think that decomposing these four levels into a 2x2 is a useful way to conceptualize their design. Additionally, our decomposition of "animation type" into a 2x2 will play a role in our statistical analysis, see sections 2.2-2.3.

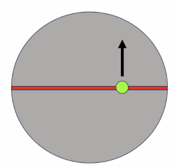
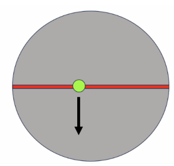
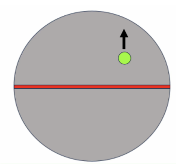
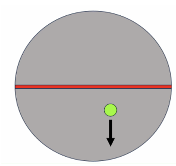
Sentence	Animation type			
	StartsRed$_{True}$		StartsRed$_{False}$	
	MovesUp$_{True}$	MovesUp$_{False}$	MovesUp$_{True}$	MovesUp$_{False}$
POS The circle wugs.				
NEG The circle does not wug.				

Table 1: Trial types in the spatial condition: four animation types crossed with two sentence types. Arrows indicate direction of movement. Note that these animations are characteristic examples; the exact starting position and amount of movement varied across the instantiations of these trial types.

Turning to the color condition, we used essentially the same design as the spatial condition but trained/tested participants on a color-based meaning for *wug*. During training, participants saw four animations of a medium-grey square becoming blacker, i.e. becoming a darker shade of grey or becoming fully black (see Figure 2).

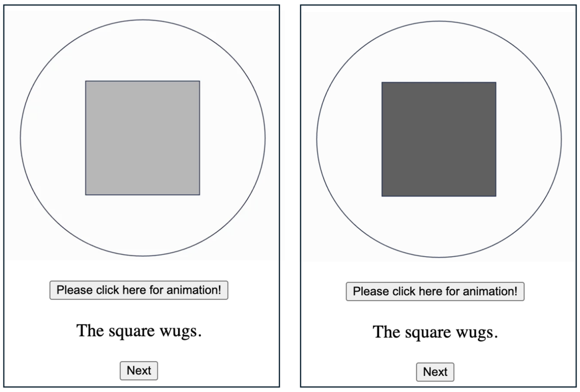


Fig. 2: Screen shots of the initial frame (left) and final frame (right) of one of the color condition's training animations.

If we follow BSC24's logic, this training should teach participants that *the square wugs* has both an initial-state entailment ("the square starts off as grey") and a change-of-state entailment ("the square darkens"). To discourage participants from assuming that *wug* has an entailment about what shade the square becomes, we had the training trials vary in how much the square darkened.

Like the spatial condition's test phase, the color condition's test phase was a covered-box task with four critical animation types that varied along two dimensions: "StartsGrey" (i.e. whether the square in the animation starts off as grey) and

“Darkens” (i.e. whether the square in the animation darkens). On “StartsGrey$_{True}$, Darkens$_{True}$” trials, the visible animation depicted an initially-grey square whose color gets closer to black (as in training). On “StartsGrey$_{True}$, Darkens$_{False}$” trials, the visible animation depicted an initially-grey square whose color gets closer to white (i.e. lightens rather than darkens). On “StartsGrey$_{False}$, Darkens$_{True}$” trials, the visible animation depicted an initially-white square that darkens (i.e. becomes grey or black). Finally, on “StartsGrey$_{False}$, Darkens$_{False}$” trials, the visible animation depicted an initially-black square that lightens (i.e. becomes grey or white).

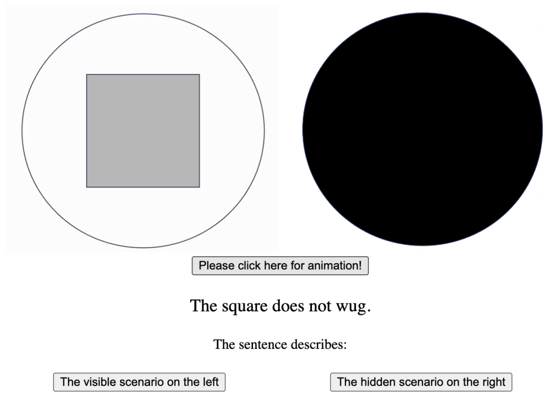


Fig. 3: Screen shot from the color condition’s test phase.

As in the spatial condition, all of these animation types occurred with positive sentences (“the square wugs”) in half of trials and occurred with negation (“the square does not wug”) in the other half, leading to the 2x2x2 design (Darkens x StartsGrey x Polarity) in Table 2. As the reader can verify, the color condition’s eight critical trial types are conceptual analogues of the spatial condition’s eight critical trial types.

In addition to critical trials, participants in the color condition saw quantified fillers (“Every/none of the circles wug”), whose visible animations were similar conceptually to the trial types listed in BSC24’s Table 5, p. 493. For example, one of the fillers in BSC24’s Experiment 1 (and thus, our spatial condition too) used an animation of four circles moving upward from the line. The analogous filler in the color condition used an animation of four initially-grey squares that all darken.

We created the animations by hand using PowerPoint and implemented the task in PCIbex (Schwarz and Zehr 2021). BSC24 coded their experiments from scratch in JavaScript (Nadine Bade, p.c.) instead of using PCIbex, so our spatial condition did not use exactly the same digital assets as their Experiment 1. Nonetheless, the stimuli in our spatial condition resembled BSC24’s Experiment 1 very closely, and as such, we do not believe that our use of PCIbex/PowerPoint affected the results.

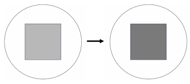
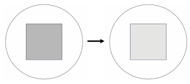
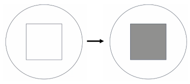
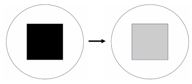
<table>
<tr><th rowspan="3">Sentence</th><th colspan="4">Animation type</th></tr>
<tr><th colspan="2">StartsGrey$_{True}$</th><th colspan="2">StartsGrey$_{False}$</th></tr>
<tr><th>Darkens$_{True}$</th><th>Darkens$_{False}$</th><th>Darkens$_{True}$</th><th>Darkens$_{False}$</th></tr>
<tr><td>POS
The square wugs.</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td>NEG
The square does not wug.</td></tr>
</table>

Table 2: Trial types in the color condition. As in Table 1, note that these animations are characteristic examples; the instantiations of these trial types varied in exactly how much the square's color changed.

2.1.3 Procedure

After providing consent, participants in both conditions saw the following short instruction text, taken directly from BSC24:480.[5]

(4) In this experiment, we will teach you the meaning of a new word *wug*. Watch the animations carefully to see how it is used.

Then, participants ran through the training phase of whichever condition they had been sorted into. After the training phase, participants saw another short text, which was taken from one of BSC24's later experiments (whose associated files were kindly sent to us by Nadine Bade, p.c.). We did not possess Experiment 1's files themselves because we did not specifically request them.

(5) Now, let's see how **you** use *wug*!

You will see two animations but only one of them is visible, the other hidden. Your task is to decide whether the sentence you will see describes the visible animation or the hidden animation. You should choose the hidden scenario if you consider the visible one inappropriate given the sentence you will see.

After reading (5), participants proceeded to the test phase of the condition they had been sorted into. The trial breakdown in the spatial condition's test phase followed the breakdown that BSC24 report for their Experiment 1 (p. 481): two "StartsRed$_{True}$, MovesUp$_{True}$" trials (one paired with a positive sentence, one paired with a negative sentence), two "StartsRed$_{False}$, MovesUp$_{False}$" trials (one positive, one negative), and four trials apiece for "StartsRed$_{True}$, MovesUp$_{False}$" and "StartsRed$_{False}$, MovesUp$_{True}$" (half positive, half negative). In addition to these 12 critical trials, participants saw one exemplar of each of the eight trial types shown in BSC24's Table 5, p. 493, leading to 20 trials total.

[5] Actually, BSC24's instructions said "...teach you a new word *wug*" instead of "...teach you the meaning of a new word *wug*." This was a minor error that we believe should not affect the results.

The trial breakdown in the color condition's test phase was parallel to the spatial condition: two "StartsGrey$_{True}$, Darkens$_{True}$" trials (one positive, one negative), two "StartsGrey$_{False}$, Darkens$_{False}$" trials (one positive, one negative), and four trials apiece for "StartsGrey$_{True}$, Darkens$_{False}$" and "StartsGrey$_{False}$, Darkens$_{True}$" (half positive, half negative). Participants also saw eight quantified fillers that were conceptual analogues of the eight trial types in BSC24's Table 5.

As in BSC24's Experiment 1, we pseudo-randomized the test-phase trial order. In the spatial condition, the first four trials were drawn from the pool of "StartsRed$_{True}$, MovesUp$_{True}$" trials and "StartsRed$_{True}$, MovesUp$_{False}$" trials; the remaining 16 trials (including the two remaining "StartsRed$_{True}$, MovesUp$_{True}$"/"StartsRed$_{True}$, MovesUp$_{False}$" trials) were presented in fully random order. Similarly, in the color condition, the first four trials were drawn from the pool of "StartsGrey$_{True}$, Darkens$_{True}$" trials and "StartsGrey$_{True}$, Darkens$_{False}$" trials, and the rest of the trials were presented in fully random order.

After a participant completed the test phase, we asked them a simple debrief question to conclude the experiment: "What did you think *wug* meant?" BSC24 did not ask their participants such a question, but we thought that doing so would be a simple and useful way to gain additional insight into participants' behavior.

2.2 Predictions

In order to replicate BSC24's Experiment 1, we would need to see the pattern shown in Figure 4 in the spatial condition.[6] Let us break down this pattern step-by-step. Starting with "StartsRed$_{True}$, MovesUp$_{True}$" trials (left panel, first column), participants should (if BSC24's results replicate) treat the visible animation on these trials as examples of "wugging," choosing the covered box at low rates for *the circle wugs* and at high rates for *the circle does not wug*. On "StartsRed$_{True}$, MovesUp$_{False}$" trials (left panel, second column), by contrast, they should treat the visible animations as examples of "not wugging," choosing the covered box at high rates for *the circle wugs* and at low rates for *the circle does not wug*. Next, participants should treat "StartsRed$_{False}$, MovesUp$_{True}$" trials (right panel, first column) as involving presupposition failure; hence, they should choose the covered box at similar rates for *the circle wugs* and *the circle does not wug* (see section 1). Finally, participants should, like BSC24's participants, treat "StartsRed$_{False}$, MovesUp$_{False}$" trials (right panel, second column) as examples of "not wugging."[7]

Statistically, the pattern in Figure 4 should yield a **three-way interaction** between StartsRed, MovesUp, and Polarity, an interaction driven by a lack of polarity contrast on "StartsRed$_{False}$, MovesUp$_{True}$" trials and a polarity contrast elsewhere.[8]

[6] BSC24 use bar graphs rather than interaction plots, but ultimately, our graphs convey the same information. We opt for interaction plots because we think that they nicely display the relevant interactions.

[7] If "starting from red" is presupposed, then one might think that participants should treat "StartsRed$_{False}$, MovesUp$_{False}$" as involving presupposition failure; however, BSC24's participants consistently treated these as examples of "not wugging." See section 3 for further discussion.

[8] As mentioned in fn. 4, BSC24 do not decompose their animation types into a 2x2. Nonetheless, we believe that decomposing "animation type" into a 2x2 and looking for a three-way interaction is a parsimonious way to probe for the predicted pattern. To confirm that BSC24's effect can indeed be described in this way, we went back to their data from Experiment 1, fit a 2x2x2 model on it (rather than the 4x2 model they fit), and found a significant three-way interaction, as expected.

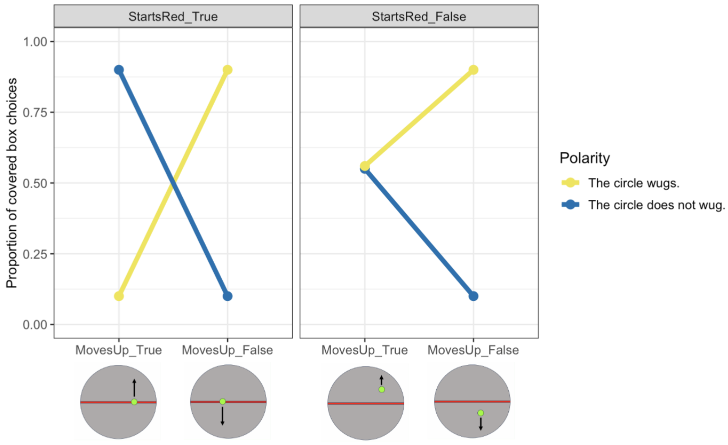


Fig. 4: Predicted response patterns if BSC24's results replicate, broken down by StartsRed (left vs. right panel), MovesUp (x-axis), and Polarity (yellow vs. blue). The examples of each animation type from Table 1 are shown under their corresponding columns in the graph.

Figure 4 is a three-way interaction because the MovesUp x Polarity interaction differs depending on whether StartsRed is "True" (left panel) or "False" (right panel).

If BSC24's results extend to the color condition, then we should see a pattern similar to that of Figure 4 on the color condition's critical trial types. Statistically, this pattern should manifest as a three-way interaction between Darkens, StartsGrey, and Polarity, an interaction driven by a lack of polarity contrast on "StartsGrey$_{False}$, Darkens$_{True}$" trials and a polarity contrast elsewhere.

As an alternative to the pattern in Figure 4, we might instead see the pattern in Figure 5, in which "StartsRed$_{False}$, MovesUp$_{True}$" trials exhibit a "not wugging" pattern (high covered-box choices for *the circle wugs*, low covered-box choices for *the circle does not wug*) rather than a presupposition-failure pattern. If we observe this pattern (or the analogous pattern in the color condition), that would suggest that most participants adopted a conjunctive construal for *wug*, i.e. they interpreted *wug* as asserting both an initial state and a change-of-state. Statistically, this pattern should also manifest as a three-way interaction, but the effect should be driven by a significant MovesUp x Polarity interaction in the "StartsRed$_{True}$" trial types and a non-significant MovesUp x Polarity interaction in the "StartsRed$_{False}$" trial types.

Finally, another pattern we could see is the one in Figure 6. The only difference between Figure 6 and the previous patterns is that "StartsRed$_{False}$, MovesUp$_{True}$" trials are treated as examples of "wugging" (low covered-box choices for *the circle wugs*, high covered-box choices for *the circle does not wug*). This pattern would suggest that most participants adopted "ignore initial state" construals, i.e. they treated *wug* as not having an entailment about the line at all (*wug* = 'moves up').

Statistically, this pattern should manifest in **the absence of a significant three-way interaction**, since the MovesUp x Polarity interaction does not differ across the

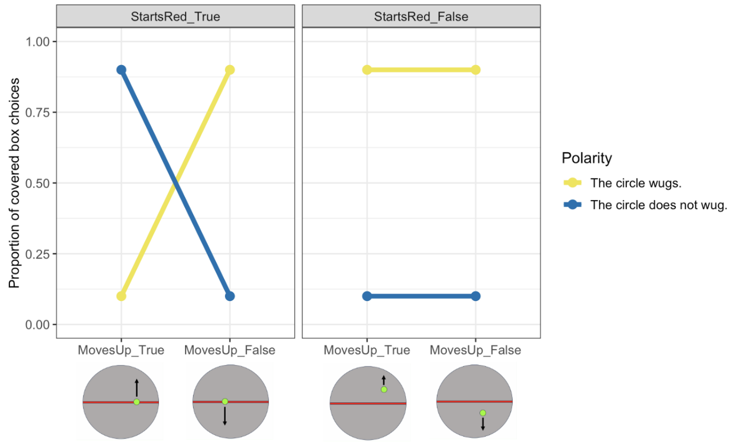


Fig. 5: Predicted response patterns if most participants adopt a conjunctive construal.

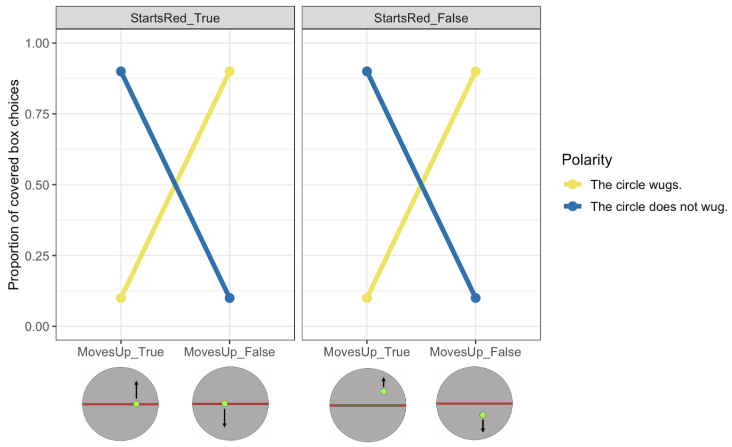


Fig. 6: Predicted pattern if most people adopt an "ignore initial state" construal.

levels of StartsRed. In other words, Figure 6's left panel has the same crossing pattern as the right panel, which would indicate that participants' behavior does not depend on whether StartsRed is "True" or "False." Note that this pattern should still yield a significant two-way interaction between MovesUp and Polarity (since the polarity effect in "MovesUp$_{True}$" trials is flipped relative to "MovesUp$_{False}$"), suggesting that participants treated upward movement as part of the assertion of *wug*.

If we observe a pattern analogous to Figure 6 in the color condition, that would indicate that a high proportion of participants treated *wug* as not having an initial-color entailment (wug = 'darkens'). Statistically, such a pattern should manifest as a significant Darkens x Polarity interaction and a non-significant three-way interaction between Darkens, StartsGrey, and Polarity.

2.3 Results

2.3.1 Spatial condition

Figure 7 shows the covered-box results in the spatial condition. Descriptively, Figure 7 resembles the pattern in Figure 6: while "MovesUp$_{True}$" trials exhibit a negation contrast that is flipped relative to "MovesUp$_{False}$" trials, the negation effect on "MovesUp$_{True}$" trials does not substantially differ across the levels of StartsRed, and neither does the negation effect on "MovesUp$_{False}$" trials.

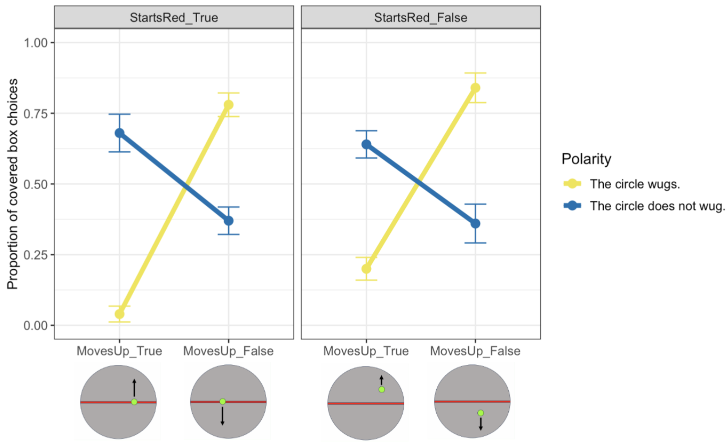


Fig. 7: Rates of covered box choices in the spatial condition, Error bars represent ±1 standard error of the mean (SEM).

To analyze these patterns statistically, we fit a mixed-effects logistic regression model on the data from the eight critical trial types, using the *glmer* function from the *lme4* package in R (Bates et al. 2015). The model predicted animation choice (visible animation vs. covered box) from MovesUp, StartsRed, Polarity, and their two- and three-way interactions. We also included by-participant and by-item random intercepts; this was the maximal random-effects structure that converged. As shown in Table 3, the model confirms what we saw in the plot: the crucial three-way interaction between MovesUp, StartsRed, and Polarity was not significant, but we did find a highly significant two-way interaction between MovesUp and Polarity.

In addition to this group-level analysis, we also performed several exploratory analyses of individual participants' behavior, the first of which looked at their responses to our debrief question: "What did you think *wug* meant?" In response to this question, 33 participants (66%) said some variant of "ascend," "rise/raise (up)," "lift," or "go up(ward)." In other words, these participants' translations for *wug* mentioned its change-of-state component but did not mention the line. An additional 8 participants (16%) mentioned the line as well as upward movement (e.g. "start at the red line and

	Estimate	Standard Error	z-value	p-value
(Intercept)	-0.61	0.31	-1.99	0.04
MovesUp	1.21	0.37	3.33	**<0.01**
StartsRed	0.05	0.36	0.12	0.9
PolarityPOS	2.34	0.49	4.73	**<0.01**
MovesUp*StartsRed	0.14	0.52	0.27	0.78
MovesUp*PolarityPOS	-4.40	0.6	-7.39	**<0.01**
StartsRed*PolarityPOS	-0.45	0.59	-0.76	0.44
MovesUp*StartsRed*PolarityPOS	-1.56	1.05	-1.48	0.13

Table 3: Summary of fixed effects from the spatial condition model.

rise," "move up from a line"). 5 more participants (10%) mentioned movement in general rather than upward movement (e.g. "when the circle move [*sic*] up or down"), and 4 participants provided uncategorizable responses (e.g. "direction").

To better understand how well participants' answers to our debrief question lined up with their covered-box choices, we visualized the results separately for each translation group (Figure 8). As shown in Figure 8a, the 33 participants whose translation mentioned upward movement but not the line displayed the same "ignore initial state" pattern that we saw in the full-sample graph (Figure 7). However, the negation contrasts in Figure 8a are even sharper than those in Figure 7. In other words, while the full sample already trended in the direction of Figure 6, we see an even stronger trend in this direction when we look at the subset visualized in Figure 8a.

As shown in Figure 8b, the 8 participants who mentioned the line strikingly displayed a conjunctive pattern rather than a presuppositional one, as evidenced by the fact that their negation contrast on "StartsRed$_{False}$, MovesUp$_{True}$" was not weaker than their negation contrast on "StartsRed$_{True}$, MovesUp$_{False}$."

Finally, Figure 8c shows that the 5 participants who mentioned movement in general rather than upward movement tended to treat every animation as an example of "wugging." Given that every animation involved the circle moving, it is unsurprising that a participant who interpreted wug as "moves" would show this pattern.[9]

To get an even finer-grained perspective on the results, we performed the same kind of analysis of individual response patterns that BSC24 ran in their Supplementary Materials (see section 3 for further discussion of their findings). The analysis into question classifies participants based on their behavior on what BSC24 call "target trials" ("StartsRed$_{True}$, MovesUp$_{False}$" trials and "StartsRed$_{False}$, MovesUp$_{True}$" trials). In particular, we categorized participants based on whether their target trials were perfectly in line with an "ignore initial state" construal, a conjunctive construal, an "ignore initial state + direction" construal (i.e. treating every animation as an example of "wugging"), or none of these construals.[10]

[9] For posterity, we also looked at the covered-box choices for the uncategorizable participants, finding that they treated positive and negative sentences exactly the same (even on "StartsRed$_{True}$, MovesUp$_{True}$" trials). We think that these participants were likely inattentive but do not exclude them *post hoc*.

[10] BSC24 did not include a "ignore initial state + direction" category in their analysis, but we think that including this category is justified by our debrief-question results. In addition, we do not include a "presuppositional" category because the exact covered-box pattern we expect to see for a presuppositional construal is not clear (we just expect a lack of strong negation contrast on "StartsRed$_{False}$, MovesUp$_{True}$")–see BSC24:483.

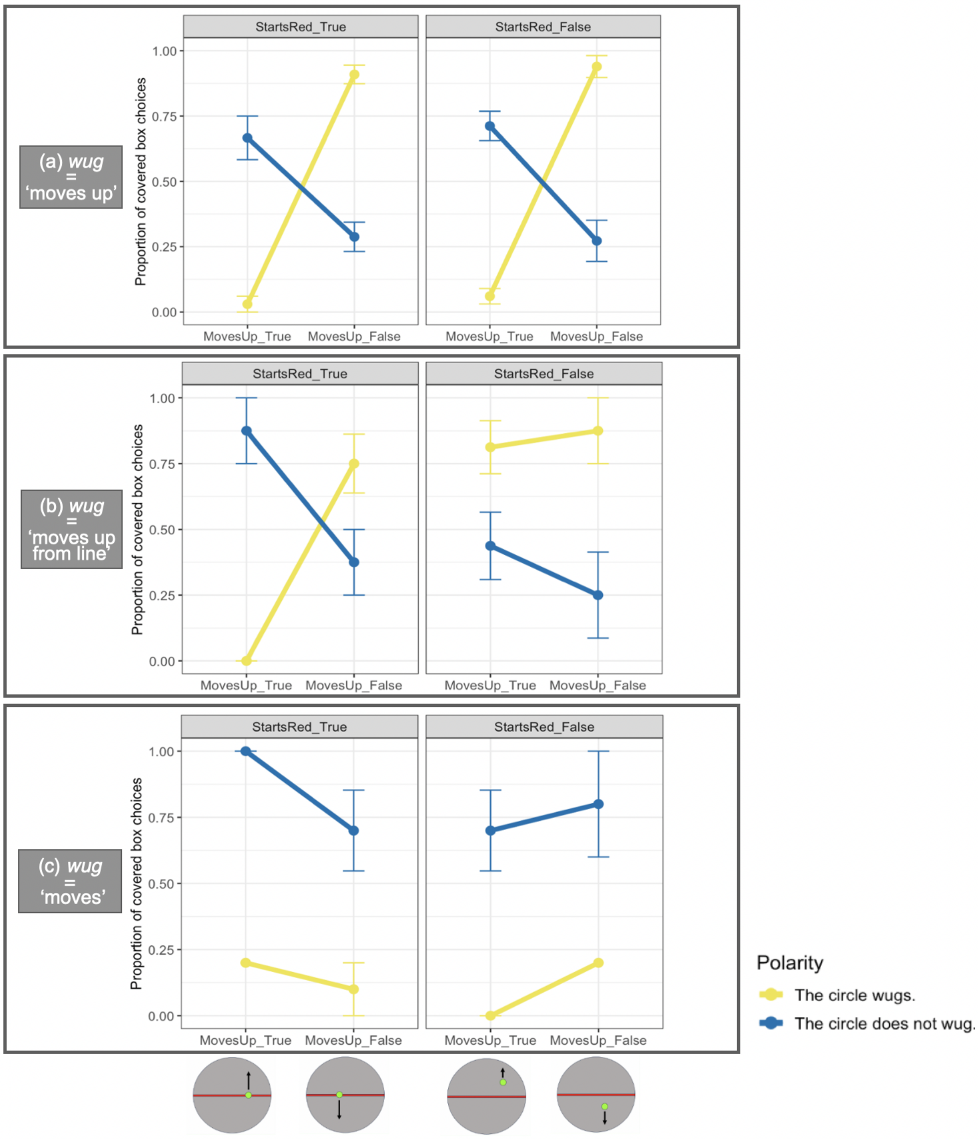


Fig. 8: Rates of choosing the covered box for each of the three major translation subgroups.

Our analysis found that 14 participants exhibited behavior on target trials that was perfectly in line with an "ignore initial state" construal. That is, on their eight target trials, these participants consistently treated "StartsRed$_{True}$, MovesUp$_{False}$" animations as examples of "not wugging" (100% covered-box choices for *the circle wugs*, 0% covered-box choices for *the circle does not wug*) and consistently treated

“StartsRed_{False}, MovesUp_{True}” animations as examples of “wugging” (0% covered-box for *the circle wugs*, 100% covered-box for *the circle does not wug*). Two more participants showed behavior that was perfectly in line with a conjunctive construal: they consistently treated both “StartsRed_{True}, MovesUp_{False}” animations and “StartsRed_{False}, MovesUp_{True}” animations as examples of “not wugging” (100% covered-box for *the circle wugs*, 0% covered-box for *the circle does not wug*). Another two participants were perfectly in line with an “ignore initial state + direction” construal, and the remaining 32 participants were not perfectly in line with any of these construals. Hence, in total, 18 participants’ (36%) target trials were perfectly in line with a non-presuppositional construal.

To gain insight into the responses of the other 32 participants, we looked at how many of their covered-box choices “leaned” towards a construal, following BSC24. We counted a participant’s responses as “leaning” towards a construal if seven of their eight target trials were in line with that construal. For example, if a participant treated “StartsRed_{True}, MovesUp_{False}” trials as “not wugging” on 3/4 trials (e.g. 100% covered-box for positive, 50% covered-box for negative) and treated “StartsRed_{False}, MovesUp_{True}” trials as “wugging” on 4/4 trials, they would be classified as leaning towards “ignore initial state.”[11] By this metric, 27 participants (54%) were perfectly in line with or leaning towards a non-presuppositional construal (21 ignore initial state, 3 conjunctive, 3 “ignore initial state + direction”).

As our final exploratory analysis, we looked at the association between a participant’s response type (i.e. what kind of behavior they displayed on target trials) and translation type (i.e. what kind of translation they offered), both when “leaning” response types are not included (Table 4a) and when they are (Table 4b). The main insight we can glean from Table 4 is that almost everyone whose behavior on target trials perfectly reflected or leaned towards a non-presuppositional construal gave a translation that was in line with that construal. For example, 20 out of the 21 participants whose responses were at least leaning “ignore initial state” gave an “ignore initial state” translation,[12] both participants in the conjunctive response group were among those who mentioned the line, etc.

[11]This criterion for “leaning” is actually more stringent than the one used by BSC24. Under BSC24’s criterion, a participant could be still be classified as leaning towards a construal even if just 2/4 of their “StartsRed_{False}, MovesUp_{True}” trials were in line with that construal. We use an “all but one” criterion, here and in section 3, in order to make our estimate of the rate of non-presuppositional construals more conservative.

[12]Note that the “odd one out” in the ignore-initial state response group may not even be an exception: their translation (“to rise from a starting position”) is ambiguous enough that it could reflect an ignore-initial-state meaning.

	Translation type				
Response type	"Wugging" is moving up	"Wugging" is moving up from the line	"Wugging" is moving	Other	**Total**
Ignore initial state	14	0	0	0	14
Conjunctive	0	2	0	0	2
Ignore initial state + direction	0	0	2	0	2
Other	19	6	3	4	32
Total	33	8	5	4	50

(a)

	Translation type				
Response type	"Wugging" is moving up	"Wugging" is moving up from the line	"Wugging" is moving	Other	**Total**
Ignore initial state, or leaning that way	20	1	0	0	21
Conjunction, or leaning that way	0	3	0	0	3
Ignore initial state + direction, or leaning that way	0	0	3	0	3
Other	13	4	2	4	23
Total	33	8	5	4	50

(b)

Table 4: Association between response type (rows) and translation type (columns) in the spatial condition. Table **(a)** shows the association when "leaning" response types are not taken into account, and Table **(b)** shows the assocation when they are.

2.3.2 Color condition

Figure 9 shows the results for the color condition. Descriptively, Figure 9 closely resembles the results of the spatial condition: "Darkens$_{True}$" trials exhibit a negation contrast that is flipped relative to "Darkens$_{False}$," but these negation effects do not vary much across the levels of StartsGrey.

Statistically, we ran into a problem known as "complete separation" when trying to fit a logistic regression model on this data. Complete separation refers to a situation in which there is no variation in the response variable for some combination of predictors. In the case of a logistic regression model whose response variable can take on the values "A" and "B," complete separation occurs when the proportion of "B"s is 0% or 100% for some combination of predictors. The data visualized in Figure 9 exhibit complete separation because there is a 0% rate of covered-box choices on "StartsGrey$_{True}$, Darkens$_{True}$, Polarity$_{POS}$" trials. As a result of this complete separation, even a model with the simplest possible random effects structure (by-participant random intercepts only) yielded a warning about abnormally large coefficients, which is a symptom of complete separation.

Complete separation can be dealt with in a number of ways, one of which is to collect more data and see if the problem goes away. Coincidentally, we happened to

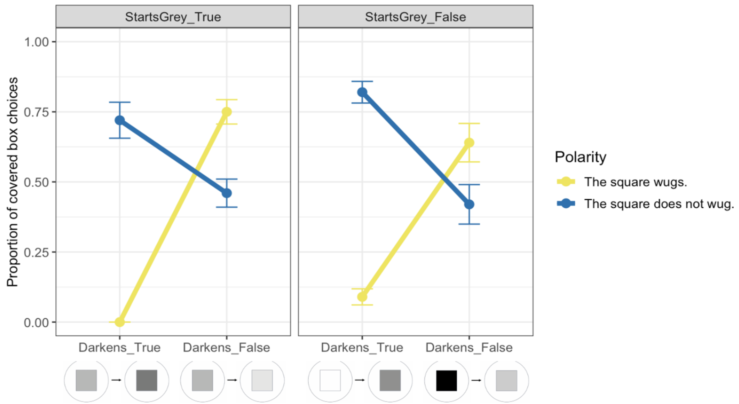


Fig. 9: Rates of covered box choices in the color condition (original 50 participants), broken down by StartsGrey, Darkens, and Polarity. The examples of each animation type from Table 2 are under their corresponding columns. Error bars are ±1 SEM.

have 15 additional "overflow" participants (participants recruited that would put us above our pre-registered sample size). We did not look at their data initially, but we include them here as a means of correcting for complete separation. No overflow participants failed our attention check, and only one had the server overload issue.

Figure 10 shows the covered-box results for the analyzed sample of 64 participants. Descriptively, Figure 10 displays a pattern similar to Figure 9, but crucially, the rate of covered-box choices on "StartsGrey$_{True}$, Darkens$_{True}$, Polarity$_{POS}$" trials is no longer 0%. Accordingly, we were now able to fit a non-pathological model. The model predicted animation choice (visible animation vs. covered-box) from Darkens, StartsGrey, Polarity, and their two- and three-way interactions. We also included by-item

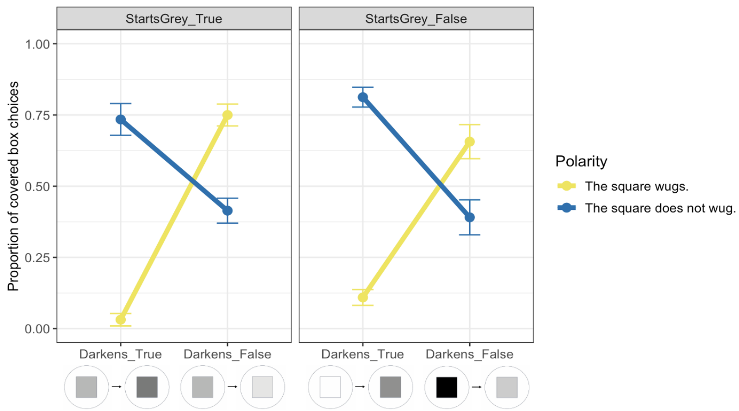


Fig. 10: Rates of covered-box choices in the color condition (analyzed sample of 64 participants). Error bars are ±1 SEM.

random intercepts as well as by-participant random intercepts and slopes for StartsGrey; this was the maximal converging random-effects structure. As shown in Table 5, there was a significant two-way interaction between Darkens and Polarity, but the crucial three-way interaction was not significant.

	Estimate	Standard Error	z-value	p-value
(Intercept)	-0.49	0.27	-1.79	0.07
Darkens	2.06	0.35	5.87	<**0.01**
StartsGrey	0.1	0.32	0.3	0.76
PolarityPOS	1.18	0.38	3.16	<**0.01**
Darkens*StartsGrey	-0.55	0.49	-1.11	0.27
Darkens*PolarityPOS	-4.99	0.53	-9.38	<**0.01**
StartsGrey*PolarityPOS	0.42	0.47	0.89	0.37
Darkens*StartsGrey*PolarityPOS	-1.39	1.02	-1.37	0.17

Table 5: Summary of fixed effects from the color condition model.

In addition to the group-level analysis whose results are shown in Table 5, we also analyzed individual behavior in the same manner as in the spatial condition. Starting with responses to our debrief question ("What did you think *wug* meant?"), 41 participants (64%) said some variant of "darkens" or "gets darker." 15 participants (23.4%) mentioned color change in general rather than darkening (e.g. "changes shade," "changes from a light colour to dark or the other way around"). Only 3 participants (4.6%) mentioned the initial-state component of *wug* (e.g. "a gradient motion from light grey to dark grey or black"), and 5 participants gave uncategorizable responses (e.g. "square").

Figure 11 visualizes the results separately for each translation group (except for the uncategorizable group). As shown in Figure 11a, the 41 participants whose translation mentioned darkening but not the initial color displayed a strong "ignore initial state" pattern, with the pairwise negation contrasts in this group being even sharper than in the full sample. As shown in Figure 11b, the 3 participants who said that *wug* has both an initial-color component and a change-of-color component exhibited a pattern that is arguably more presuppositional than conjunctive, but the small sample size makes it difficult to draw conclusions. Finally, Figure 11c shows that the 15 participants who provided translations like "changes shade" tended to treat every animation as "wugging," which makes sense given that every animation involved a color change.

Next, we investigated individual response patterns in the same manner as in the spatial condition. We found that 18 participants exhibited behavior on their eight target trials that was perfectly in line with an "ignore initial state" construal. That is, these participants consistently treated "StartsGrey$_{True}$, Darkens$_{False}$" trials as examples of "not wugging" (100% covered-box for *the square wugs*, 0% covered-box for *the square does not wug*) and consistently treated "StartsGrey$_{False}$, Darkens$_{True}$" trials as "wugging" (0% covered-box for *the square wugs*, 100% covered-box for *the square does not wug*). No participants' target trials were perfectly in line with a conjunctive construal, 2 participants' target trials were perfectly in line with a "color change"

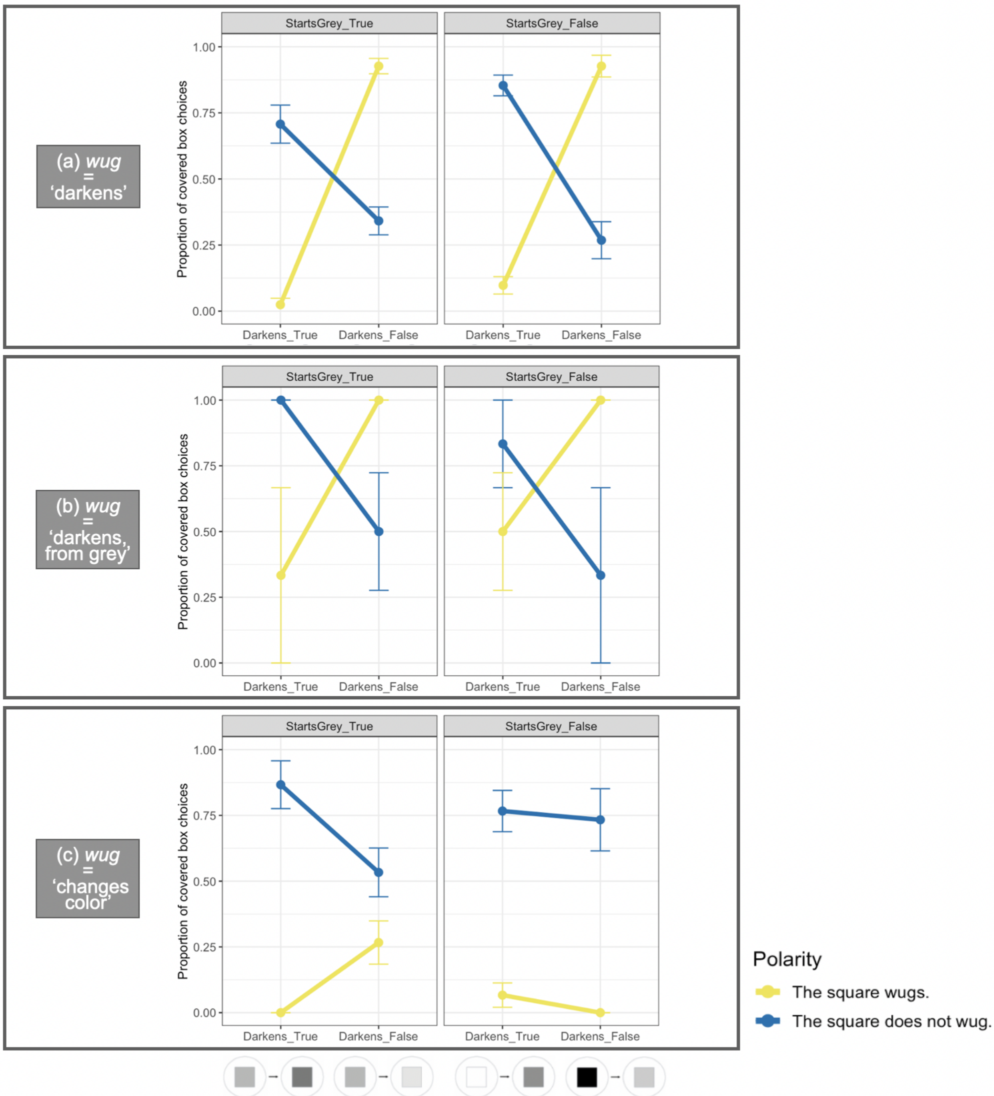


Fig. 11: Rates of choosing the covered box for the three major translation subgroups, color condition.

construal (i.e. treating every animation as an example of "wugging"), and the remaining 44 participants were not perfectly in line with any of these construals. To gain insight into these 44 participants' behavior, we looked at how many of these participants' covered-box choices "leaned" towards a construal, where "leaning" was defined as in the spatial condition (7/8 target trials in line with a construal). By this metric, 42 participants were perfectly in line with or leaning towards a non-presuppositional construal (32 "ignore initial state", 1 conjunctive, 9 "color change").

	Translation type				
Response type	"Wugging" is darkening	"Wugging" is darkening, starting from grey	"Wugging" is changing color	Other	**Total**
Ignore initial state	16	1	1	0	18
Conjunctive	0	0	0	0	0
Color change	0	0	2	0	2
Other	25	2	12	5	44
Total	41	3	15	5	64

(a)

	Translation type				
Response type	"Wugging" is darkening	"Wugging" is darkening, starting from grey	"Wugging" is changing color	Other	**Total**
Ignore initial state, or leaning that way	29	1	1	1	32
Conjunction, or leaning that way	1	0	0	0	1
Color change, or leaning that way	1	0	7	1	9
Other	10	2	7	3	22
Total	41	3	15	5	64

(b)

Table 6: Association between response type (rows) and translation type (columns) in the color condition. Table **(a)** shows the association when "leaning" response types are not taken into account, and Table **(b)** shows the assocation when they are.

Table 6 shows the association between translation type and response type in the color condition, both when "leaning" participants are not taken into account (Table 6a) and when they are (Table 6b). As in the spatial condition, we see that the vast majority of people whose behavior on target trials perfectly reflected or leaned towards a non-presuppositional construal also offered a translation for *wug* that was in line with that construal. The pattern in Table 6 is slightly noisier than the corresponding pattern in the spatial condition (Table 4), however: that is, a handful of participants in Table 6 whose target trials were suggestive of a non-presuppositional construal gave an incongruous translation for *wug*. Overall, though, the association between response type and translation type trends in the same direction in our two conditions.

2.4 Discussion

In our experiment, we tested the validity of BSC24's's paradigm in two ways. First, in our spatial condition, we tried to replicate their Experiment 1; second, in our color condition, we sought to ascertain whether the results of their Experiment 1 generalize to a nonce change-of-state predicate other than the one that they test.

Neither the results of the spatial condition nor the results of the color condition are in line with what BSC24 would expect. Starting with the color condition, the group-level results were not consistent with presuppositional construals being dominant, and in fact, they indicate that “ignore initial state” construals were much more common. Additionally, the group-level results in the spatial condition were the same as the color condition, which constitutes a replication failure vis-à-vis BSC24. Taken together, these results cast doubt on the robustness of BSC24’s paradigm: the color condition’s results suggest that BSC24’s original effect does not straightforwardly generalize to all kinds of change-of-state predicates, and the spatial condition’s results suggest that even their original effect is fragile.

Our analyses of individual response patterns confirmed what our group-level analyses already led us to suspect: namely, that “ignore initial state” construals (*wug* = “move up” in the spatial condition, *wug* = “darkens” in the color condition) were highly prevalent. In addition, these analyses revealed that other types of non-presuppositional construals were fairly common as well. For example, a sizable group of participants ignored not only the initial-state component of *wug* but also the directionality of the change-of-state, interpreting *wug* as “moves” in the spatial condition or interpreting *wug* as “changes color” in the color condition. Furthermore, in the spatial condition at least, we found evidence that non-presuppositional construals were common even among the minority of participants who seemed not to ignore the initial state; in particular, conjunctive construals appeared to be common among this group. Overall, the prevalence of non-presuppositional construals in our experiment chafes against the idea that participants in BSC24’s paradigm exhibit an overwhelming bias towards presuppositional construals.

Finally, our analyses of participants’ responses to our debrief question (“What did you think *wug* meant?”) revealed a strong association between the type of translation a participant offered and their covered-box choices. For one thing, visualizing the results separately based on translation subgroup revealed that different translation subgroups had starkly different patterns of covered-box choices, patterns that mirrored their translations. Additionally, among those whose covered-box choices were (almost) perfectly in line with a certain construal, virtually all of them provided a translation for *wug* that was in line with that construal.

The correlation between translation type and response type was not perfect in the sense that some participants’ covered-box choices were not consistently in line with any construal (and were thus not aligned with their translation); nonetheless, we believe that the correlation is strong enough to lend credence to a worry originally raised by BSC24’s reviewers. In particular, BSC24 at one point discuss a reviewer comment concerning the possibility that participants in their task are consciously translating *wug* (p. 499). As they note, it would be problematic for the logic of their experiments if conscious translation of *wug* were common, since in that case, one can no longer be sure that the results of their experiments reflect conceptual biases rather than linguistic biases or knowledge. Although BSC24 push back on the hypothesis that their participants were translating *wug* in a few ways (e.g. they question why participants would overwhelmingly translate *wug* in such a way that “upward” is focused/at-issue rather than “red line”), they acknowledge that further research is needed to assess the role of translation in how people approach their task. To a certain extent, our debrief-question results can be thought of as addressing the role of translation in BSC24’s paradigm. In particular, the observed association between translation type and response type suggests that conscious translation may indeed be common, which in turn would cast doubt on their crucial assumption that their paradigm probes conceptual rather than linguistic biases.

In summary, our results do not support the conceptual-bias hypothesis. At the same time, however, our results do not necessarily disprove the conceptual-bias hypothesis either; rather, they suggest that artificial word learning paradigms are not a robust way to identify the relevant conceptual biases, if they in fact exist. To elaborate, the most salient aspect of our results (namely, the high rate of "ignore initial state" construals) does not adjudicate between conceptual and contextual approaches; rather, the availability of this simple strategy suggests that BSC24's paradigm is, as it stands, not a robust way to probe how participants encode initial states (since participants can just ignore the initial state). The one aspect of our results that arguably bears on the conceptual vs. contextual debate is the fact that construing *wug* as conjunctive appears to be an available strategy, at least in our spatial condition. If we accept BSC24's linking hypothesis, conjunctive construals being available would be problematic for a conceptual-bias view (such as BSC24's own) according to which the human brain has a "triggering algorithm" that always encodes initial states as presuppositions. However, given that the conjunctive group is relatively small, it may be too hasty to draw strong conclusions on the basis of this group, and hence, this is not a knock-down argument against the conceptual-bias hypothesis.

3 Non-presuppositional construals in BSC24's experiments

At this point, the main question that remains is: why were non-presuppositional construals of *wug* so prevalent in our experiment, in a way that they (seemingly) were not in BSC24's? In response, we argue that there is reason to suspect that a substantial proportion of participants in BSC24's experiments–up to half–in fact adopted either an "ignore initial state" construal or a conjunctive construal. To preview our argument, "ignore initial state" construals were widespread among a subset of participants in BSC24's Experiment 2; while the authors suggest that this result reflects a quirk of how these participants were trained, we argue on the basis of a closer look at the authors' Supplementary Materials that non-presuppositional construals were common even among participants who were trained differently. The upshot of our argument is that our results constitute less of a departure from BSC24's original experiments than it would seem at first glance: non-presuppositional construals were widespread not just in our experiment but in BSC24's original experiments too, although the ratio of "ignore initial state" construals to conjunctive construals was lower in (some of) their experiments than in ours.

As the first step in our argument, we take a closer look at BSC24's Experiment 2 (section 3.1), the experiment of theirs in which non-presuppositional construals undeniably played a role. Then, we turn to non-presuppositional construals in their Experiment 1 (section 3.2), the other experiment they consider in their Supplementary Materials. Having argued that non-presuppositional construals were widespread based on a granular look at individual experiments, we conclude this section by noting that our re-interpretation of the data has an additional empirical advantage over BSC24's, one that cuts across experiments (section 3.3).

3.1 Experiment 2

As mentioned in section 1, BSC24's Experiment 2 probes whether the covered-box results of Experiment 1 are robust across different training types, i.e. whether the results of Experiment 1 replicate even when participants are trained differently. To that end, BSC24 split participants in Experiment 2 into two groups. Both groups were told in training that *the green circle wugs* describes upward movement from the line (as in Experiment 1). Unlike in Experiment 1, however, participants in Experiment 2 also received training with negation: the "canonical" training group was taught that *the green circle does not wug* describes downward movement from the line, while the "non-canonical" training group was taught that *the green circle does not wug* describes upward movement that does not start from the line.

"Canonical" training was included as a sanity check, since there is no reason *prima facie* to hypothesize that such training would change the results of Experiment 1. "Non-canonical" training, by contrast, was intended to bias participants against construing the initial-state component of *wug* as presuppositional. If participants in the non-canonical group construe "starts on the line" as presuppositional in spite of this bias, BSC24 argue, then that would suggest that there is a conceptual bias to encode initial states as presuppositions, one that cannot be easily overriden. In sum, BSC24 predict that "irrespective of training, participants should still construe the initial state as presuppositional" (p. 488).

Contrary to this prediction, the results from the two training groups did significantly differ: the non-canonical group exhibited the same pattern as Experiment 1's participants (akin to Figure 4), while the canonical group displayed an "ignore initial state" pattern akin to Figure 6. This finding about the canonical group conflicts with BSC24's predictions. In response to this conflict, they briefly suggest that the canonical group's results may reflect "the special pragmatics of training with negation" (p. 490); they exhort the reader to instead focus on the non-canonical group's results, which they interpret as evidence that "non-canonical" training cannot override the conceptual bias to encode initial states as presuppositions.

While BSC24 suggest that the prevalence of "ignore initial state" construals in the canonical group reflects a quirk of how this group was trained, we believe that a closer look at BSC24's Supplementary Materials provides reason to suspect that non-presuppositional construals of *wug* were available much more generally. In particular, BSC24's Supplementary Materials contain details about individual participants' response patterns in Experiments 1 and 2 (as already noted in section 2.3), and it is these patterns that lead us to suspect that non-presuppositional construals were widespread, even outside of Experiment 2's canonical training group.

For Experiment 2, they found that out of the 37 participants in the canonical group, 18 (48.6%) exhibited behavior on target trials that was perfectly in line with an "ignore initial state" interpretation. One other participant showed behavior that was perfectly in line with a conjunctive interpretation, and the remaining participants did not perfectly match up with an "ignore initial state" construal or a conjunctive one.

These individual-level patterns are fairly unsurprising in the sense that they confirm what BSC24's analysis on the full dataset already led us to suspect: namely, that "ignore initial state" construals were highly prevalent in the canonical training group.

The individual-level results from the non-canonical group, by contrast, provide additional insight above and beyond the group-level analysis and, in our estimation, speak against BSC24's interpretation of the non-canonical group's results. The crucial finding is that out of the 33 participants in the non-canonical group, 16 (48.4%) displayed behavior on target trials that was perfectly in line with either an "ignore initial state" construal (9 participants) or a conjunctive construal (7 participants). These individual-level patterns suggest that despite appearances, the proportion of participants who adopted non-presuppositional construals in the non-canonical group is similar to the proportion who adopted such construals in the canonical group; the only difference is that non-presuppositional construals in the non-canonical group were split between "ignore initial state" and conjunction, while non-presuppositional construals in the canonical group were mostly "ignore initial state." What this means is that that the prevalence of non-presuppositional construals was visible in the full sample for only the canonical group because in the non-canonical group, the "ignore initial state" participants and the conjunctive ones canceled each other out. In other words, even though 16 out of 33 participants in the non-canonical group exhibited a categorical negation contrast on "StartsRed$_{False}$, MovesUp$_{True}$" trials, the 9 "ignore initial state" participants' negation contrast on these trials (0% covered-box for *the circle wugs*, 100% for *the circle does not wug*) was flipped relative to the 7 conjunctive participants (100% covered-box for *the circle wugs*, 0% for *the circle does not wug*), leading to a non-significant negation contrast in the full sample.

The fact that nearly half of participants clearly adopted a non-presuppositional construal already goes against BSC24's claim (p. 490) that the non-canonical group's results reflect an overwhelming bias to construe initial states as presuppositions. But beyond this, even for the 17 participants in the non-canonical group who were not consistently "ignore initial state" or conjunctive, there is reason to suspect that not all of them adopted the construal that BSC24 claim is dominant (i.e. upward movement = assertion, starting from line = presupposition). For example, 11 out of these 17 participants exhibited behavior on "StartsRed$_{True}$, MovesUp$_{False}$" trials that is unexpected if they were construing upward movement as part of the assertion of *wug* (e.g. they chose the covered box on "StartsRed$_{True}$, MovesUp$_{False}$" regardless of the polarity of the sentence).[13] This finding casts further doubt on the idea that the non-canonical group's results reflect a prevalent bias to construe changes-of-state as assertive and initial-states as presupposed.

3.2 Experiment 1

Taking stock so far, non-presuppositional construals of *wug* were widespread in both training groups in BSC24's Experiment 2. But were they also widespread in their Experiment 1? Well, if we look at the "strict" version of their individual-level analysis

[13] More specifically, 6 of these 17 participants exhibited the negation contrast on "StartsRed$_{True}$, MovesUp$_{False}$" trials that we expect if upward movement is part of the assertion, 2 had the opposite negation contrast, 6 chose the covered box regardless of polarity, and 3 chose the visible animation regardless of polarity.

on Experiment 1 (i.e. the version that does not take "leaning" response types into account), we find that only 13 out of 49 participants (26.5%) displayed behavior on target trials that was perfectly in line with an "ignore initial state" interpretation (9 participants) or a conjunctive interpretation (4 participants). This metric suggests that relative to Experiment 2's participants, a substantially lower proportion of Experiment 1's participants adopted non-presuppositional construals, although a rate of 26.5% is arguably still high enough to be worrisome.

However, the version of the analysis that includes "leaning" response types warrants suspicion that the rate of non-presuppositional construals may have been much higher than 26.5%. In particular, the crucial finding is that 24 out of 49 participants (48.9%) in Experiment 1 were perfectly in line with or leaning towards a non-presuppositional construal (17 "ignore initial state", 7 conjunctive), where a participant is counted as "leaning" towards a construal if all but one of their target trials were in line with that construal. As in Experiment 2's non-canonical group, we can deduce that the high proportion of (leaning) non-presuppositional participants in Experiment 1 was not apparent in the full sample because on "StartsRed$_{False}$, MovesUp$_{True}$" trials, the influence of those who were (leaning) "ignore initial state" was largely mitigated by the influence of those who were (leaning) conjunctive.

Overall, these results suggest that non-presuppositional construals were common in BSC24's experiments, as in ours, although this was not immediately visible in the experiments exhibiting BSC24's original effect because the "ignore initial state" participants in these experiments were counter-balanced by a sizable group of conjunctive participants.

3.3 An argument from "StartsRed$_{False}$, MovesUp$_{False}$" trials

To close this section, we would like to note that this perspective on the findings helps explain an otherwise puzzling aspect of the BSC24 data, namely the results for "StartsRed$_{False}$, MovesUp$_{False}$" trials. As discussed in section 2.2, participants in all of BSC24's experiments (and in ours) tended to treat the visible animations on these trials as examples of "not wugging," with high covered-box choices for *the circle wugs* and low covered-box choices for *the circle does not wug*. However, this result is actually puzzling for the hypothesis that most participants treated *wug* as presuppositional (see fn. 7). In particular, if "starting from the line" were overwhelmingly encoded as a presupposition, then one would expect that participants should treat "StartsRed$_{False}$, MovesUp$_{False}$" trials as examples of presupposition failure (i.e. there should be no difference between "StartsRed$_{False}$, MovesUp$_{False}$" and "StartsRed$_{False}$, MovesUp$_{True}$"), but they did not. In response to this issue, BSC24 suggest that local accommodation may be at work in "StartsRed$_{False}$, MovesUp$_{False}$" trials (p. 486), but then the question arises of why local accommodation was unavailable for "StartsRed$_{False}$, MovesUp$_{True}$" trials; they do not address this question in detail.

While BSC24's findings about "StartsRed$_{False}$, MovesUp$_{False}$" trials (which we replicated) are problematic for their interpretation of the results, they are completely unproblematic under our re-interpretation, according to which non-presuppositional construals (specifically, "ignore initial state" and conjunctive) were widespread. To see this, note that under both an "ignore initial state" construal and a conjunctive one, the visible animations on "StartsRed$_{False}$, MovesUp$_{False}$" should be examples of "not wugging." While the two construals should yield the same negation contrast on "StartsRed$_{False}$, MovesUp$_{False}$," they should manifest in opposite negation contrasts on "StartsRed$_{False}$, MovesUp$_{True}$" trials (as explained above). In sum, our re-interpretation of BSC24's data predicts the observed contrast between "StartsRed$_{False}$, MovesUp$_{True}$" trials and "StartsRed$_{False}$, MovesUp$_{False}$" trials without further assumptions, whereas their interpretation of the data does not.

4 General discussion

When adults encounter a novel word against the backdrop of a change-of-state event, do they automatically analyze it as presupposing the initial state and asserting the change of state, thereby revealing a productive triggering mechanism that generalizes beyond their own lexicon? Our findings point to a negative answer. This is not to deny that there are genuine cognitive biases underlying presupposition triggering. Rather, our claim is that artificial-word-learning paradigms are simply poor probes for detecting them.

The relevant problems are two-fold, and they cut to the core of the linking assumptions. First, participants in these artificial word-learning studies are adults with a full-fledged linguistic system, and our evidence suggests they leaned heavily on that system when doing the task. Rather than mapping nonce words to new concepts, they seem to have simply been translating from one language to another. This is perhaps unsurprising: natural language is one of the most efficient tools at an adult's disposal for parsing a scene, and the close match between participants' spontaneous translations and their behavioral responses shows that they were giving the scene a linguistic description in their own language and then assigning a meaning based on that description. If so, the behavior we observe cannot speak to productive triggering mechanisms, nor underlying conceptual biases.

Second, even if there are conceptual biases that favor certain presuppositional profiles, these biases may not express themselves in artificial paradigms because the setup — where participants have to map word meanings from exposure to a scene — is too far removed from how word learning actually happens. The problem, famously pointed out by Quine (1960) and brought to sharp relief in Gleitman's (1990) seminal work on verb learning, is that the extralinguistic context in which a word is used massively underdetermines its meaning. The same scene is compatible with multiple descriptions, sometimes necessarily so. In our study, and BSC24's, the depicted event was intended as a "move up from the red line" event. But in actuality, any event where an object "moves up from the red line" is also a "moving" event, a "moving up" event, a "starting from line" event, and so on. Participants in our study exploited this indeterminacy fully, mapping the nonce verb onto many different meanings–all of which were consistent with training.

The problem of learning via observing a scene is especially challenging for backgrounded content. To begin, learners must notice it. Why should the learner pay attention to backgrounded information at all? And even if they do, how do they distinguish the backgrounded content that is encoded by a word, from the plethora of backgrounded content that is not part of its meaning? Our results speak to these challenges, suggesting that when learning from observation, people tend to ignore content that they consider less salient altogether (rather than encoding it as a presupposition).

This problem of accounting for which items are presupposition triggers and what kind of presuppositions they trigger remains one of the central challenges for semanticists. Yet learners solve the problem, and it seems conceivable to us that some bias is needed to get help them with the task. Yet identifying such biases and finding empirical evidence for them is anything but straightforward. We do not claim to have solved the problem, only that artificial learning paradigms do not move us any closer.

Data availability

Our stimuli, dataset and analysis script can be found at this link, and our preregistration is available at this link.

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